

GLENWOOD, MN, USA

WMO: 726547

Lat: 45.644N

Lon: 95.320W

Elev: 1393

StdP: 13.97

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04931

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-11.4	-24.0	1.5	-16.6	-18.0	2.2	-11.0	31.3	10.9	27.5	13.7	8.4	290	0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	89.7	72.9	85.8	71.4	82.3	69.0	76.5	84.7	74.5	82.2	72.4	79.6	13.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.4	130.7	81.1	72.2	125.6	80.1	70.2	116.9	77.7	40.9	84.7	38.9	82.6	37.1	79.5	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.4	24.5	21.3	DB	-23.4	94.3	6.4	2.9	-28.0	96.4	-31.8	98.1	-35.3	99.7	-40.0	101.9	(4)
(5)			WB	-23.7	79.9	6.1	2.1	-28.1	81.4	-31.7	82.6	-35.1	83.8	-39.6	85.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.4	11.3	15.8	28.6	43.7	56.5	66.5	71.3	68.7	61.0	46.7	31.4	17.6		
	DBStd	23.30	13.85	13.19	12.83	10.35	8.77	6.45	5.73	5.47	8.30	9.39	11.26	12.27		
	HDD50	4860	1201	958	671	237	38	0	0	0	9	176	564	1006		
	HDD65	8422	1666	1378	1130	640	288	57	14	26	173	570	1009	1471		
	CDD50	2449	0	0	6	48	239	496	660	580	339	75	6	0		
	CDD65	533	0	0	0	1	24	102	209	140	53	4	0	0		
	CDH74	4704	0	0	0	27	306	891	1832	1133	463	51	1	0		
	CDH80	1379	0	0	0	7	92	240	598	308	123	11	0	0		
	WSAvg	10.1	10.4	10.5	10.8	11.7	11.5	9.5	8.1	8.0	9.4	10.6	10.7	10.2		
	PrecAvg	26.3	0.6	0.6	1.3	2.3	2.9	4.2	3.9	3.3	3.1	2.5	1.1	0.6		
(15) Precipitation	PrecMax	36.3	1.9	1.3	3.4	6.4	5.4	7.5	8.0	7.4	5.7	6.6	4.0	1.6		
	PrecMin	18.5	0.0	0.1	0.4	0.7	0.3	1.9	0.8	0.4	1.0	0.4	0.0	0.1		
	PrecStd	4.2	0.4	0.3	0.8	1.4	1.2	1.5	1.7	1.6	1.5	1.7	1.0	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.7	49.9	66.3	77.2	89.4	90.6	93.4	90.9	89.6	81.1	65.7	46.1		
		MCWB	36.8	46.2	55.5	57.5	65.9	73.2	75.5	73.6	70.8	62.0	54.6	40.2		
	2%	DB	36.3	42.6	57.4	71.8	81.7	86.0	89.9	86.3	83.5	72.4	57.1	39.4		
		MCWB	33.9	38.4	50.2	54.4	63.0	70.7	74.6	72.1	69.5	59.2	47.3	35.8		
	5%	DB	33.8	36.9	51.9	65.7	76.5	82.0	86.0	82.4	79.8	65.8	52.0	36.3		
		MCWB	32.3	34.7	46.4	51.4	59.8	67.7	73.2	69.8	66.5	56.1	45.8	34.1		
	10%	DB	29.8	34.3	45.3	61.2	72.2	79.3	82.3	81.0	75.0	61.3	46.0	33.9		
		MCWB	29.4	32.3	41.1	48.8	58.4	65.9	70.6	68.8	64.7	53.3	41.4	32.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.3	45.1	57.3	61.1	70.8	76.9	80.2	78.2	75.1	66.9	57.2	41.9		
		MCDB	40.6	49.0	62.3	71.4	80.9	86.1	89.1	86.9	83.3	74.2	63.4	46.1		
	2%	WB	34.1	38.3	51.7	56.8	66.8	74.0	77.6	75.0	72.1	61.7	50.3	36.9		
		MCDB	36.0	41.1	56.9	67.3	75.6	82.0	85.7	82.5	79.6	68.6	56.6	39.6		
	5%	WB	31.9	35.3	45.8	53.2	63.5	71.1	75.4	72.9	69.6	57.4	45.7	34.3		
		MCDB	33.3	37.1	50.3	62.4	71.2	78.5	83.2	79.9	76.2	64.9	50.3	35.9		
	10%	WB	29.0	32.3	41.1	50.1	60.2	68.4	73.3	70.7	66.9	54.2	41.8	32.0		
		MCDB	29.4	34.1	45.6	59.2	69.1	76.3	80.4	77.4	73.2	60.1	46.7	33.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.2	17.0	16.6	20.0	20.5	19.4	19.8	19.9	20.3	17.9	15.4	14.3		
		MCDBR	17.2	17.6	24.3	29.5	28.1	23.7	23.2	23.0	25.1	26.3	23.5	17.7		
		MCWBR	14.8	14.6	15.9	15.7	13.4	11.5	11.2	11.4	12.6	14.4	15.2	14.4		
		MCDBR	16.3	16.8	22.8	26.1	23.2	19.9	21.0	20.5	21.0	22.7	21.5	16.0		
	5% WB	MCWBR	14.3	14.1	15.8	16.1	13.8	12.2	12.3	12.4	12.8	14.6	15.3	13.4		
(40) Clear-Sky Solar Irradiance	taub		0.261	0.280	0.307	0.357	0.383	0.384	0.382	0.390	0.356	0.309	0.277	0.253		
	taud		2.415	2.359	2.404	2.333	2.318	2.369	2.374	2.343	2.423	2.539	2.531	2.456		
	Ebn at Noon		274	287	294	286	280	280	279	272	273	275	267	266		
	Edh at Noon		22	29	32	37	39	38	37	37	31	24	20	19		
(44) All-Sky Solar Radiation	RadAvg		542	889	1206	1499	1714	1899	2032	1695	1292	814	518	377		
	RadStd		69	100	110	126	131	125	74	141	97	118	50	49		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page